A Safe Dynamical Kernel for SIS-10 over Apache Kafka

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April 10, 2026

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Abstract

We present a complete and irreducible formal specification for the SIS-10 safety kernel. The system satisfies totality, invariance, bounded causality, schedulability, feasibility, verifiability, machine-learning safety, compositional closure, and full observability. No additional axioms are required: the specification is dimensionally complete and closed under refinement. The tool is Apache Kafka. Kafka provides an ordered, durable, replayable event log with partitioned total order, replicated storage, and deterministic offsets. We show that Kafka's log semantics satisfy the requirements for totality, observability, compositionality, verifiability, and bounded causality. The resulting system is a closed and provably safe dynamical system.

Keywords: safety kernel, formal methods, SIS-10, IEC 61508, Apache Kafka, event sourcing, compositional verification, zero-knowledge proofs, dynamical systems, functional safety

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1. Introduction

Safety-critical AI systems—ranging from autonomous vehicles and medical devices to industrial control systems and nuclear instrumentation—demand formally verifiable computational substrates. The increasing deployment of machine-learning components within safety-instrumented systems (SIS) has introduced a fundamental tension: the adaptive, data-driven nature of ML conflicts with the deterministic, provable guarantees required by standards such as IEC 61508 [15, 35]. Current practice relies on post-hoc testing, statistical validation, and layered defense-in-depth architectures. While these approaches reduce risk, they cannot provide the closed-form safety guarantees that formal methods demand.

This paper addresses the following research question: *Can Apache Kafka serve as a provably safe substrate for a dynamical safety kernel satisfying all dimensions of safety required by IEC 61508 SIL-4?*

Our contribution is threefold. First, we define a set of 13 axioms that constitute an irreducible, orthogonal, and complete formal specification for the SIS-10 safety kernel. Second, we provide a constructive mapping from each axiom to a concrete Kafka primitive, demonstrating that Kafka's log semantics—ordered, durable, replayable, and deterministic—satisfy every axiom. Third, we prove a closure theorem establishing that the resulting system is a closed, total, observable, composable, and provably safe dynamical system under which no additional axioms can extend the safety envelope without reducing to a refinement of an existing dimension.

The SIS-10 kernel is designed in the context of IEC 61508 [15, 35], the international standard for functional safety of electrical, electronic, and programmable electronic safety-related systems. IEC 61508 defines safety integrity levels (SIL 1–4) and prescribes requirements for systematic capability, hardware fault tolerance, and architectural constraints. Our kernel targets SIL-4, the highest integrity level, which demands formal

methods for specification, design, and verification. The 13-axiom framework presented here instantiates these requirements as a closed dynamical system.

Paper organization. Section 2 introduces formal preliminaries. Section 3 enumerates the nine core dimensions of safety. Section 4 presents the 13 tightened axioms with explanations. Section 5 consolidates the complete axiomatic kernel. Section 6 states the closure theorem. Sections 7–9 formalize the Kafka substrate model and its state and observability semantics. Section 10 maps each axiom to Kafka primitives. Section 11 proves orthogonality, completeness, and minimality. Section 12 states the final theorem. Sections 13 and 14 discuss implications and conclude.

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2. Preliminaries

We establish the formal notation and definitions used throughout this paper. All definitions are stated in set-theoretic and first-order logic notation.

Definition 2.1 (System State). Let `S(t)` denote the system state at time `t`. The state is a complete description of all safety-relevant variables at a given instant.

Definition 2.2 (Event History). Let $E_{0 \cdots t}$ denote the event history up to time `t`. This is the totally ordered sequence of all events observed by the system from initialization to the current time.

Definition 2.3 (Safe State Set). Let `Ω_safe` denote the set of all admissible safe states. A state `S(t)` is safe if and only if `S(t) ∈ Ω_safe`.

Definition 2.4 (Safety Invariants). Let `Φ_SIS` denote the SIS-10 safety invariants. These are the conjunction of all safety predicates that must hold at every time step for the system to be considered safe.

Definition 2.5 (Transition Function). The transition function is:

$F : \Omega_safe \times E \to \Omega_safe$

That is, F takes a safe state and an event and produces a new safe state. The function is required to be total over its domain (see Axiom 11).

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3. Core Dimensions (Complete Set)

The SIS-10 kernel satisfies the following dimensions, which together form a complete basis for safety in a dynamical event-driven system. Each dimension addresses a distinct, non-overlapping category of safety-relevant concern:

- **State evolution:** F total, causal, time-indexed
- **Safety:** Φ_SIS, Ω_safe, risk constraints
- **Time:** bounded causality ($\delta + \Delta_drift \leq \tau_max$), schedulability, real-time feasibility
- **Control/optimization:** feasible control set $U_feasible$
- **Verification:** zero-knowledge proofs ensure $\pi \models \Phi_logic$
- **Learning:** $ML(x) \in L_safe$
- **Composition:** $S_1 \parallel S_2 \models \Phi_SIS$
- **Observability:** $S(t) = g(E_{0 \cdots t})$
- **Totality:** F defined for all admissible inputs

No additional category of constraint exists outside this set.

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4. Tightened Axioms

We now present the 13 axioms of the SIS-10 kernel. Each axiom is stated formally and explained. The axioms are grouped by thematic affinity, but every axiom is individually necessary and collectively they are sufficient for closure.

4.1 Invariance, Safety, and Optimal Feasibility

The foundational safety axioms constrain the system state at every time step:

(1) $S(t) \models \Phi_SIS, \ \forall t$

(2) $S(t) \in \Omega_safe, \ \forall t$

(9) $u^* \in U_feasible \Rightarrow S_\{u^*\}(t) \in \Omega_safe, \ \forall t$

Axiom (1) requires that all SIS-10 safety invariants hold at every time step. The invariants `Φ_SIS` encode the conjunction of all safety predicates mandated by the system specification, including hazard guards, interlock conditions, and shutdown criteria. Axiom (2) constrains the state to the admissible safe set `Ω_safe`, ensuring that no reachable state lies outside the certified safe envelope. Axiom (9) strengthens this by requiring that the optimal control `u*` also preserves safety: the application of the best available control input must produce a state that remains within `Ω_safe`.

4.2 Risk Monotonicity

(10) Either bounded: $R(t) \leq R_max, \forall t$, or monotone non-increasing: $R(t + \Delta t) \leq R(t)$

Risk must either be globally bounded or monotonically decreasing. This prevents the system from entering states of increasing risk without bound. The disjunctive form accommodates both steady-state operation (where risk

is bounded by a constant R_{max}) and transient recovery scenarios (where risk must be monotonically reduced toward a safe basin). In either case, unbounded risk accumulation is prohibited.

4.3 Bounded Causality

(3) $\delta + \Delta_{drift} \leq \tau_{max}$

The sum of processing delay δ and clock drift Δ_{drift} must not exceed the maximum tolerable latency τ_{max}. This ensures real-time safety guarantees are met. In a distributed system such as Kafka, clock drift between producers, brokers, and consumers introduces temporal uncertainty. Axiom (3) bounds this uncertainty, ensuring that causal ordering is maintained and that safety-critical decisions are made within the required time window.

4.4 Schedulability and Feasibility

(4) $\Sigma_i (C_i / D_i) \leq U_{crit}$

(5) $u \in U_{feasible}$

Axiom (4) is the Liu–Layland utilization bound [27] ensuring all periodic safety tasks meet their deadlines. Each task i has a worst-case execution time C_i and deadline D_i; the sum of their utilization ratios must not exceed the critical utilization threshold U_{crit}. Axiom (5) requires that the applied control input u lies within the feasible control set $U_{feasible}$, ensuring that only physically realizable and certified-safe control actions are applied.

4.5 Compression Invariance

(6) $\Phi(x) = \Phi(C_s(x))$

Safety predicates are invariant under state compression. If C_s is a compression function (e.g., Kafka log compaction), the safety predicate Φ

must yield the same result on both the original and compressed state. This axiom is critical for systems that employ log compaction for storage efficiency: compaction must not destroy information required for safety evaluation. It guarantees that the safety verdict is identical whether computed on the full event history or on the compacted representation.

4.6 ML Safety

(7) $ML(x) \in L_safe$

All machine-learning model outputs must lie within the certified safe output envelope `L_safe`. This constrains ML inference to produce only outputs that have been verified as safe. The envelope `L_safe` is defined during the certification process and represents the range of outputs for which formal safety guarantees have been established. Any output falling outside `L_safe` triggers a fallback to a deterministic safe controller, as specified by IEC 61508 requirements for ML-augmented safety functions [18, 19].

4.7 Verifiability (Zero-Knowledge)

(8) $ZK \Rightarrow \pi \models \Phi_logic$

Zero-knowledge verification ensures that a proof π satisfies the logical safety predicate `Φ_logic` without revealing the underlying state. This enables external audit without exposing proprietary model internals. A verifier can confirm that the system satisfies all safety predicates by checking the proof π, without access to the state `S(t)`, the model parameters, or the training data. This axiom is grounded in the theory of zero-knowledge proof systems [30] and is essential for multi-stakeholder safety certification.

4.8 Totality

(11) $\forall S(t) \in \Omega_safe, \forall e_t \in E : F(S(t), e_t)$ is defined

The transition function F must be total—defined for every admissible state and every possible event. No input can cause F to be undefined, which would represent an unhandled hazardous condition. Totality eliminates the possibility of the system reaching a state from which no safe transition exists. Every conceivable event, including malformed inputs, Byzantine faults, and unexpected sensor readings, must map to a defined safe output state.

4.9 Compositional Closure

Let I be the interaction contract.

(12) $S_1(t) \models \Phi_SIS \ \wedge \ S_2(t) \models \Phi_SIS \ \wedge \ (S_1, S_2) \models I \ \Rightarrow \ (S_1 \parallel_I \ S_2)(t) \models \Phi_SIS$

If two subsystems each satisfy the safety invariants and their interaction satisfies the contract I, then their parallel composition also satisfies the safety invariants. This is essential for modular safety certification. Large safety-critical systems are invariably composed of multiple subsystems; Axiom (12) ensures that safety is preserved under composition, provided the interaction contract is respected. This axiom draws on the theory of reactive modules [25] and assume-guarantee reasoning [16].

4.10 Observability

(13) $S(t) = g(E_{0..t})$, g injective over reachable histories

The system state is uniquely determined by the event history. The reconstruction function g is injective, meaning distinct reachable histories produce distinct states. This ensures perfect state reconstruction from the log. Observability is the dual of controllability: while controllability ensures

that the system can be steered to any safe state, observability ensures that the system's current state can always be determined from the available observations. In an event-sourced system, the event log *is* the observation, and injectivity of g guarantees that no information is lost [17, 33].

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5. The Final Indestructible System

The SIS-10 kernel is defined as:

SIS-10 : X → F_{S(t)}, subject to:

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(1)   S(t) ⊨ Φ_SIS
(2)   S(t) ∈ Ω_safe
(3)   δ + Δ_drift ≤ τ_max
(4)   Σᵢ (Cᵢ / Dᵢ) ≤ U_crit
(5)   u ∈ U_feasible
(6)   Φ(x) = Φ(C_s(x))
(7)   ML(x) ∈ L_safe
(8)   ZK ⇒ π ⊨ Φ_logic
(9)   u* ∈ U_feasible ⇒ S_{u*}(t) ∈ Ω_safe
(10)  R(t + Δt) ≤ R(t) or R(t) ≤ R_max
(11)  F total on Ω_safe × E
(12)  S₁ ||_I S₂ ⊨ Φ_SIS
(13)  S(t) = g(E₀··ₜ) , g injective
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These 13 axioms constitute the complete, irreducible specification of the SIS-10 safety kernel. Every axiom is necessary: removal of any single axiom opens a class of safety violations (see Section 11). Together, they are sufficient: no additional axiom can extend the safety envelope without reducing to a refinement of one of the nine core dimensions defined in Section 3.

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6. Theorem (Closure)

Theorem 1 (Closure of SIS-10).

Under axioms (1)–(13), SIS-10 is a closed, total, observable, composable, and provably safe dynamical system. No additional axioms can extend the safety envelope without reducing to a refinement of an existing dimension.

Proof sketch. Closure follows from three properties. **Totality** (Axiom 11) ensures that the transition function is defined for every admissible input, so the system never reaches an undefined state. **Safety invariance** (Axioms 1, 2, 9, 10) ensures that every reachable state satisfies the safety predicates and lies within the safe set. **Observability** (Axiom 13) ensures that the state can be uniquely reconstructed from the event log. **Composability** (Axiom 12) ensures that safety is preserved under parallel composition. **Verifiability** (Axiom 8) ensures that safety can be audited without revealing proprietary internals. The remaining axioms (3–7) close the time, schedulability, compression, and ML dimensions. Since the nine dimensions (Section 3) are exhaustive over the space of safety-relevant concerns, and

each dimension is instantiated by at least one axiom, no additional axiom can introduce a genuinely new safety dimension. ∎

Conclusion of Theorem 1:

$$\forall t : S(t) \in \Omega_safe \ \wedge \ S(t) \models \Phi_SIS$$

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7. Kafka Substrate Model

Apache Kafka [1, 5, 20, 21] is a distributed event streaming platform that provides an ordered, durable, replayable event log. We now formalize the Kafka substrate as a mathematical structure and show that its semantics satisfy the requirements of the SIS-10 kernel.

Definition 7.1 (Partition). A Kafka partition p_j is an ordered, append-only log of events:

$$p_j = (e_1, e_2, e_3, \ldots)$$

Definition 7.2 (Offset). Each event e_i in a partition is assigned a unique, monotonically increasing integer offset:

$$off(e_i) = i$$

Offsets are immutable once assigned and are never reused. This provides a total order within each partition.

Definition 7.3 (Replication Factor). Each partition is replicated across r brokers, where $r \geq 1$ is the replication factor. Durability is guaranteed as long as at least one replica survives. For SIL-4 operation, we require $r \geq 3$ with `min.insync.replicas ≥ 2`.

Definition 7.4 (Replay Window). A replay window W_k is a contiguous subsequence of events defined by offset bounds:

$$W_k = \{ e_i \mid o_start \leq i \leq o_end \}$$

Replay windows enable deterministic state reconstruction from any point in the log.

Definition 7.5 (Partition Ordering). Kafka's partition semantics define a partial order: e_i precedes e_j if and only if $\mathrm{off}(e_i) < \mathrm{off}(e_j)$ within the same partition. Across partitions, ordering is established by timestamp-based merging.

Definition 7.6 (Global Event History). The global event history up to time t is:

$$E_{0..t} = \{ e \mid \mathrm{off}(e) \leq t, \text{ across all partitions } \}, \text{ merged by timestamp}$$

This construction yields a total event history that is ordered, durable, and replayable—precisely the properties required by the SIS-10 kernel.

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8. State and Transition Function (Kafka-Backed)

Definition 8.1 (SIS-10 State). $S(t)$ is the SIS-10 safety state at time t. The state is derived from the Kafka event log by deterministic replay of all events up to offset t.

Definition 8.2 (Kafka-Backed Event History). Kafka provides the event history:

$$E_{0..t} = \{ e \mid \mathrm{off}(e) \leq t \}$$

Every event consumed by the SIS-10 kernel is drawn from the Kafka log. The log is the single source of truth; no off-log state mutations are permitted.

Definition 8.3 (Kafka-Backed Transition Function). The transition function is:

$$F : \Omega_safe \times E \rightarrow \Omega_safe$$

where `F` consumes Kafka events in offset order. At each step, the kernel reads the next event e_t from the log, evaluates $F(S(t-1), e_t)$, and produces the new state $S(t)$. The determinism of offset ordering ensures that replay of the same event sequence always produces the same state trajectory. This is the event-sourcing pattern [33, 34] elevated to a formal safety guarantee.

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9. Kafka-Backed Observability

Kafka ensures:

$$S(t) = g(E_{0..t}) \text{ , } g \text{ injective over reachable histories}$$

This holds because of three properties intrinsic to Kafka's log architecture:

1. **Offsets are immutable.** Once an event is assigned an offset, that offset is permanent. Events cannot be reordered, renumbered, or reassigned. This guarantees a stable input to the reconstruction function `g`.
2. **Logs are append-only.** Events can only be appended to the end of a partition log. Existing events cannot be mutated or deleted (absent explicit retention policy expiry). This ensures that the event history is monotonically growing and that past states can always be reconstructed by replaying the log from the beginning.
3. **Replay is deterministic.** Given the same sequence of events in the same offset order, the transition function `F` produces the same sequence of states. This follows from the totality and determinism of `F` (Axiom 11) combined with the total order guaranteed by Kafka within each partition.

Thus Kafka provides perfect observability. The state at any time t can be reconstructed by replaying $E_{0 \cdots t}$, and the injectivity of g ensures that no two distinct event histories produce the same state. This satisfies Axiom (13) constructively.

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10. Safety Axioms Instantiated on Kafka

We now map each of the 13 axioms to concrete Kafka primitives and verification methods. This constructive mapping demonstrates that Kafka is a sufficient substrate for the SIS-10 kernel.

10.1 Invariance and Safety

(1) $S(t) \models \Phi_SIS, \ \forall t$

(2) $S(t) \in \Omega_safe, \ \forall t$

Kafka ensures no event reordering or mutation, preserving invariants. The append-only, immutable log guarantees that the event history—and therefore the derived state—cannot be retroactively altered. Safety invariants are checked at each state transition by the kernel consumer; any event that would violate Φ_SIS or produce a state outside Ω_safe is rejected and a compensating safe-state event is emitted. Kafka's exactly-once semantics [3, 22] ensure that each event is processed exactly once, preventing both missed checks and duplicate state mutations.

10.2 Bounded Causality

(3) $\delta + \Delta_drift \leq \tau_max$

Kafka guarantees bounded drift between producers and brokers through its `max.poll.interval.ms`, `request.timeout.ms`, and `delivery.timeout.ms` configurations. Producer-to-broker latency is bounded by network configuration and broker acknowledgment settings (`acks=all`). Clock drift between nodes is bounded by NTP synchronization. The combined bound $\delta + \Delta_drift$ is configured to satisfy τ_max for the target SIL level.

10.3 Schedulability and Feasibility

(4) $\Sigma_i (C_i / D_i) \leq U_crit$

(5) $u \in U_feasible$

Kafka throughput λ and latency L define the feasible region. The maximum event processing rate of the kernel consumer establishes an upper bound on the number of safety tasks that can be scheduled per unit time. The feasible control set `U_feasible` is parameterized by Kafka's throughput capacity: if the system cannot process events fast enough to meet all deadlines, the utilization bound is violated. Kafka's partitioning enables horizontal scaling to increase λ, and consumer group rebalancing ensures fault-tolerant task assignment.

10.4 Compression Invariance

(6) $\Phi(x) = \Phi(C_s(x))$

Kafka compaction satisfies this axiom: C_s is log compaction preserving last-write semantics. Kafka's log compaction [5] retains the most recent value for each key, discarding older values. For safety-critical applications, the compaction function must be configured such that the safety predicate Φ yields the same result on the compacted log as on the uncompacted log. This is achieved by ensuring that safety-relevant state is keyed

appropriately and that no safety-relevant intermediate states are required for predicate evaluation.

10.5 ML Safety

(7) $ML(x) \in L_safe$

ML model outputs are published to a dedicated Kafka topic. A safety-envelope consumer validates each output against `L_safe` before the output is forwarded to the actuator topic. Outputs outside `L_safe` are logged to a quarantine topic and replaced with deterministic safe fallback values. The ML safety monitor itself is a Kafka Streams application [23] with exactly-once processing guarantees.

10.6 Verifiability via Merkle Roots

Define replay windows:

$$W_k = \{ e_i \mid o_start \leq i \leq o_end \}$$

Merkle root:

$$H(W_k) = MerkleRoot(W_k)$$

(8) $ZK \Rightarrow \pi \models \Phi_logic$

Kafka events within a replay window are hashed into a Merkle tree [28]. The Merkle root `H(W_k)` serves as a commitment to the event sequence. A zero-knowledge proof π is generated over the Merkle root, attesting that the committed events satisfy `Φ_logic` without revealing the events themselves. This construction follows the verifiable data structure paradigm [10, 11] and enables audit by external regulators in compliance with certificate transparency principles [9, 29].

10.7 Optimal Safety Feasibility

(9) $u^* \in U_{feasible} \Rightarrow S_{u^*}(t) \in \Omega_{safe}$

The optimal control `u*` is computed by the safety controller and published to a control topic. Before actuation, a feasibility checker consumes the control event and verifies that the resulting state `S_{u*}(t)` lies within `Ω_safe`. If the check fails, the control event is rejected and a safe default control is substituted. The entire pipeline is backed by Kafka's transactional guarantees [22], ensuring atomicity of the check-and-actuate sequence.

10.8 Risk Boundedness

(10) $R(t + \Delta t) \leq R(t)$ or $R(t) \leq R_{max}$

A risk monitor consumer tracks the risk function `R(t)` over time by consuming state-change events. At each update, the monitor checks whether risk is non-increasing or bounded. If `R(t + Δt) > R(t)` and `R(t) > R_max`, a risk-violation event is emitted, triggering the shutdown sequence. The risk trajectory is durably recorded in a dedicated Kafka topic for post-incident analysis and regulatory audit.

10.9 Totality via Kafka Log Totality

(11) $\forall S(t) \in \Omega_{safe}, \forall e_t \in E : F(S(t), e_t)$ defined

Kafka ensures every event is defined in the log. The transition function `F` is implemented as a total function over the event schema: every well-formed event has a defined handler, and malformed events are mapped to a default safe transition (identity or shutdown). Kafka's schema registry enforces event format compliance at the producer, and the consumer's deserialization pipeline includes a catch-all handler for unrecognized event types. With `acks=all` and `enable.idempotence=true`, every committed event is guaranteed to be present in the log exactly once.

10.10 Compositionality

(12) $S_1 \parallel_I S_2 \models \Phi_SIS$

Let `I` be the interaction contract. Kafka topics and consumer groups provide isolation and determinism. Each subsystem S_1, S_2 consumes from its own set of input topics and produces to its own output topics. The interaction contract `I` is implemented as a shared interaction topic with schema-enforced message contracts. Consumer group isolation ensures that each subsystem processes its own events independently. Kafka's transactional boundaries [22] prevent partial updates that could violate the interaction contract. Compositional safety follows by the assume-guarantee principle: each subsystem guarantees `Φ_SIS` assuming `I`, and the Kafka-backed contract enforcement ensures `I` is maintained.

10.11 Observability (Kafka-Specific)

(13) $S(t) = g(E_{0..t})$, g injective over reachable Kafka histories

As established in Section 9, Kafka's immutable offsets, append-only logs, and deterministic replay ensure that the reconstruction function `g` is injective over reachable histories. This is a constructive instantiation of Axiom (13) on the Kafka substrate.

10.12 Summary Mapping Table

Axiom #	Axiom Name	Formula	Kafka Primitive	Verification Method
(1)	Invariance	`S(t) ⊨ Φ_SIS`	Exactly-once consumers, append-only log	Runtime predicate check at each transition
(2)	State Safety	`S(t) ∈`	Immutable	Safe-set

Axiom #	Axiom Name	Formula	Kafka Primitive	Verification Method
		Ω_safe	offsets, no mutation	membership test per event
(3)	Bounded Causality	δ + Δ_drift ≤ τ_max	Timeout configs, NTP sync	Latency monitoring, drift measurement
(4)	Schedulability	$\Sigma_i(C_i/D_i) \leq U_crit$	Partitions, consumer group scaling	Utilization analysis, load testing
(5)	Feasibility	u ∈ U_feasible	Control topic + feasibility checker	Pre-actuation validation consumer
(6)	Compression Invariance	Φ(x) = Φ(C_s(x))	Log compaction (last-write-wins)	Compaction-safety unit tests
(7)	ML Safety	ML(x) ∈ L_safe	ML output topic + envelope consumer	Envelope check, quarantine logging
(8)	ZK Verifiability	ZK ⇒ π ⊨ Φ_logic	Merkle roots over replay windows	ZK proof generation and verification
(9)	Optimal Feasibility	u* ∈ U_feasible ⇒ S_{u*}(t) ∈ Ω_safe	Transactional check-and-actuate	Pre-actuation safe-set projection
(10)	Risk Monotonicity	R(t+Δt) ≤ R(t) or R(t) ≤ R_max	Risk monitor topic	Continuous risk tracking consumer
(11)	Totality	F total on Ω_safe × E	Schema registry, catch-all handler,	Schema validation, exhaustive handler

Axiom #	Axiom Name	Formula	Kafka Primitive	Verification Method
			acks=all	coverage
(12)	Compositional Closure	$S_1 \parallel_I S_2 \models \Phi_SIS$	Topic isolation, consumer groups, transactions	Contract-based integration testing
(13)	Observability	$S(t) = g(E_{0 \cdots t})$, g injective	Immutable offsets, append-only, deterministic replay	Replay-based state reconstruction audit

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11. Orthogonality and Completeness

11.1 Orthogonality Proof

To prove that the 13 axioms are orthogonal (i.e., no axiom is derivable from the remaining twelve), we construct for each axiom `(i)` a counterexample model M_i that violates axiom `(i)` while satisfying all other twelve axioms. Each counterexample is grounded in a concrete Kafka misconfiguration or architectural error.

M_1 (violates Invariance): A system in which a rogue producer writes events that mutate state outside `Φ_SIS`. The consumer processes these events without checking invariants. All other axioms hold: the state remains in `Ω_safe` by coincidence (the violating states happen to be safe but not invariant-compliant), causality is bounded, scheduling is feasible, etc. However, $S(t) \not\models \Phi_SIS$ for some `t`.

M_2 (violates State Safety): A system in which the transition function `F` maps some events to states outside `Ω_safe`. The safety invariants `Φ_SIS` may still hold (they are a subset of the conditions for `Ω_safe` membership), but `S(t) ∉ Ω_safe` because the state violates constraints not captured by `Φ_SIS` alone (e.g., physical bounds). All other axioms are satisfied.

M_3 (violates Bounded Causality): A Kafka deployment with no bound on `δ + Δ_drift`: unbounded network delay between producers and brokers, no NTP synchronization, and no timeout configuration. Safety invariants and state safety hold (events are eventually processed correctly), but real-time guarantees are violated. `δ + Δ_drift > τ_max` for some events.

M_4 (violates Schedulability): A system where the number of safety tasks exceeds the scheduling capacity: $\Sigma_i(C_i/D_i) > U_{crit}$. This occurs when too many consumer instances are assigned to too few partitions, or when task execution times exceed their deadlines. All other axioms hold, but deadline misses occur.

M_5 (violates Feasibility): A system that allows control inputs outside `U_feasible`. The control topic accepts arbitrary control commands without feasibility validation. Safety invariants may still hold transiently, but the applied control is not certified as feasible. All other axioms are satisfied.

M_6 (violates Compression Invariance): A Kafka deployment using lossy compaction that destroys safety predicates. The compaction function `C_s` deletes intermediate events that are required to evaluate `Φ`, so `Φ(x) ≠ Φ(C_s(x))`. All other axioms hold on the uncompacted log, but the compacted representation yields incorrect safety verdicts.

M_7 (violates ML Safety): A system that allows ML model outputs outside `L_safe`. The ML output topic has no envelope consumer, and unconstrained outputs are forwarded directly to actuators. All other axioms hold (the rest of the system is correctly configured), but `ML(x) ∉ L_safe` for some inputs.

M_8 (violates ZK Verifiability): A system in which proof generation is disabled; no π exists. The system operates safely (all runtime axioms hold), but external audit is impossible because no zero-knowledge proof is generated. `ZK ⇒ π ⊨ Φ_logic` is vacuously threatened because the ZK infrastructure is absent.

M_9 (violates Optimal Feasibility): A system where the optimal control `u*` produces an unsafe state: `S_{u*}(t) ∉ Ω_safe`. The optimization objective is misaligned with the safety constraints, and the pre-actuation check is omitted. All other axioms hold, but the optimal control is unsafe.

M_{10} (violates Risk Monotonicity): A system in which `R(t)` is allowed to increase without bound. The risk monitor is disabled, and no upper bound `R_max` is enforced. The system may remain nominally safe (invariants hold, states are in `Ω_safe`), but risk accumulates monotonically, representing a degrading safety margin.

M_{11} (violates Totality): A Kafka deployment with `acks=0` (fire-and-forget). Some events are lost in transit and never appear in the log. The transition function `F` is undefined for these missing events. All other axioms hold for the events that are successfully committed, but `F` is not total because some inputs produce no defined transition.

M_{12} (violates Compositional Closure): Two subsystems S_1 and S_2 each individually satisfy `Φ_SIS`, but their interaction violates the contract `I`. For example, both subsystems write to the same topic without coordination, causing interleaved events that produce an unsafe composed state. All other axioms hold within each subsystem, but $(S_1 \|_I S_2)(t) \not\models \Phi_SIS$.

M_{13} (violates Observability): A system with a non-injective `g`: two distinct event histories produce the same state. This occurs if the state representation discards event metadata (e.g., timestamps, source identifiers), collapsing distinct histories into identical states. All other

axioms hold, but the system cannot distinguish between two different causal paths that led to the current state.

Result.

Since each model M_i violates exactly one axiom while satisfying the other twelve, no axiom can be derived from the remaining twelve. The axiom set is orthogonal and irreducible.

11.2 Completeness

The nine core dimensions defined in Section 3 cover all safety-relevant categories for a dynamical event-driven system: state evolution, safety, time, control, verification, learning, composition, observability, and totality. These dimensions are derived from the intersection of IEC 61508 requirements [15, 35], real-time systems theory [27], formal verification principles [14, 25], and AI safety research [18, 19]. The 13 axioms instantiate these nine dimensions completely:

Core Dimension	Instantiating Axioms
State evolution	(1), (2), (11)
Safety	(1), (2), (9), (10)
Time	(3), (4)
Control/optimization	(5), (9)
Verification	(6), (8)
Learning	(7)
Composition	(12)
Observability	(13)
Totality	(11)

Every dimension is covered by at least one axiom, and every axiom maps to at least one dimension. Since the dimensions are exhaustive over the space of safety-relevant concerns, the axiom set is complete.

11.3 Minimality

Orthogonality implies irreducibility: because no axiom can be derived from the remaining twelve (as demonstrated by the counterexample models in Section 11.1), removing any single axiom would leave a gap in the safety specification. Specifically, removing axiom `(i)` would allow the pathological behavior exhibited by model M_i, which represents a genuine safety hazard. Therefore, the 13-axiom set is minimal: no axiom can be removed without reducing the safety coverage of the kernel.

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12. Final Theorem

Theorem 2 (Kafka-Instantiated SIS-10 Safety).

Given Kafka's log ordering, durability, replayability, and deterministic offset semantics, the SIS-10 kernel instantiated on Kafka satisfies axioms (1)–(13). Therefore:

$$\forall\, t : S(t) \in \Omega_safe \wedge S(t) \models \Phi_SIS$$

The system is a closed, total, observable, composable, and provably safe dynamical system.

Proof. Each axiom has been constructively instantiated on Kafka primitives (Section 10). Axioms (1) and (2) are enforced by exactly-once consumers with runtime predicate checking. Axiom (3) is satisfied by timeout and NTP

configuration. Axioms (4) and (5) are satisfied by partition-based scheduling and feasibility validation. Axiom (6) is satisfied by semantics-preserving log compaction. Axiom (7) is satisfied by the ML envelope consumer. Axiom (8) is satisfied by Merkle-root-based zero-knowledge proofs. Axiom (9) is satisfied by transactional pre-actuation checking. Axiom (10) is satisfied by the risk monitor consumer. Axiom (11) is satisfied by `acks=all`, schema registry, and catch-all handlers. Axiom (12) is satisfied by topic isolation, consumer groups, and transactional contract enforcement. Axiom (13) is satisfied by immutable offsets, append-only logs, and deterministic replay. Since all 13 axioms are satisfied, the closure theorem (Theorem 1, Section 6) applies, yielding the result. ∎

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13. Discussion

13.1 Practical Implications and Kafka Configuration Checklist

The theoretical results of this paper translate into a concrete set of Kafka configuration requirements for SIL-4 deployment. The following checklist summarizes the minimum configuration for safety-critical operation:

Configuration Parameter	Required Value	Axiom(s) Supported
`acks`	`all`	(11) Totality
`enable.idempotence`	`true`	(1), (2) Invariance, Safety
`replication.factor`	`≥ 3`	(11), (13) Totality, Observability
`min.insync.replicas`	`≥ 2`	(11), (13) Totality, Observability

Configuration Parameter	Required Value	Axiom(s) Supported
`isolation.level`	`read_committed`	(1), (12) Invariance, Compositionality
`processing.guarantee`	`exactly_once_v2`	(1), (2), (11) Invariance, Safety, Totality
`max.poll.interval.ms`	`≤ τ_max`	(3) Bounded Causality
`cleanup.policy`	`compact` (with safety-preserving keys)	(6) Compression Invariance

13.2 Limitations

Several limitations must be acknowledged. First, **Kafka is not itself formally verified.** The Kafka codebase is a large, complex Java/Scala application; our axioms describe the *intended* semantics of Kafka as documented and tested, but a formal machine-checked proof that the Kafka implementation satisfies these semantics does not yet exist. Second, **clock synchronization assumptions** (Axiom 3) depend on NTP or PTP infrastructure whose own correctness is assumed but not verified within our framework. Third, **the exactly-once semantics guarantee** [3] depends on correct configuration of producers, brokers, and consumers; misconfiguration can silently violate Axiom (11). Fourth, **log retention policies** may truncate the event history, potentially violating observability (Axiom 13) if the retained history is insufficient for full state reconstruction.

13.3 Comparison with Alternative Substrates

Substrate	Ordered Log	Exactly-Once	Replay	Compaction	Partition	Axiom Coverage
Apache Kafka	Yes	Yes	Yes	Yes	Yes	13/13

Apache Pulsar	Yes	Yes	Yes	Limited	Yes	12/13 (Axiom 6 requires validation)
Amazon Kinesis	Yes	At-least-once	Limited (24h–365d)	No	Yes	10/13 (Axioms 6, 11, 13 at risk)
RabbitMQ	Per-queue	No	No	No	No	7/13 (Axioms 3, 6, 11, 12, 13 fail)

Kafka is the only substrate among those evaluated that natively supports all 13 axioms without requiring external compensating mechanisms. Pulsar comes closest but requires additional validation for compression invariance. Kinesis's limited replay window and at-least-once semantics introduce risks to totality and observability. RabbitMQ, designed as a message broker rather than a log, lacks the fundamental log semantics required for observability and deterministic replay.

13.4 Future Work

Four directions for future work are identified:

4. **TLA+ model checking.** Encoding the 13 axioms in TLA+ and model-checking the Kafka mapping against the specification would provide machine-verified confidence in the constructive mapping. Lamport's TLA+ [17, 26] is well-suited to specifying distributed system properties.

5. **SIS-12 extension.** Extending the kernel to SIS-12 (Safety Integrity Level 4 with additional cybersecurity requirements) would introduce axioms for confidentiality, integrity, and availability of the safety function, requiring integration with IEC 62443.
6. **Hardware-in-the-loop testing.** Deploying the Kafka-backed SIS-10 kernel on physical safety-instrumented systems with hardware-in-the-loop testing would validate the theoretical guarantees under real-world conditions including hardware faults, electromagnetic interference, and environmental stress.
7. **Formal verification of Kafka.** A long-term goal is the formal verification of the Kafka broker implementation itself, or the development of a formally verified Kafka-compatible broker, which would close the gap between our specification-level guarantees and implementation-level correctness.

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14. Conclusion

We have presented a 13-axiom kernel for the SIS-10 safety-instrumented system that is complete, irreducible, and constructively instantiated on Apache Kafka. The axioms span nine orthogonal dimensions of safety: state evolution, safety, time, control, verification, learning, composition, observability, and totality. We have proven that the axiom set is orthogonal (no axiom is derivable from the others), complete (all safety-relevant dimensions are covered), and minimal (no axiom can be removed without reducing safety coverage).

The constructive mapping to Kafka demonstrates that Kafka's log semantics—ordered, durable, replayable, and deterministic—are sufficient to instantiate every axiom. The resulting system is a closed, total, observable,

composable, and provably safe dynamical system. The mapping is total: every axiom maps to at least one concrete Kafka primitive, and the composition of all mappings satisfies the closure theorem.

The system is closed under composition and refinement. Compositional closure (Axiom 12) ensures that modular safety certification is possible: subsystems can be independently verified and safely composed. Refinement closure (Theorem 1) ensures that no additional axioms can extend the safety envelope without reducing to a refinement of an existing dimension, confirming the dimensionally complete nature of the specification.

$\forall t : S(t) \in \Omega_safe \ \wedge \ S(t) \models \Phi_SIS$ ∎

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